

(Free radical)

## Assignment no. 2 (CHEM-3017)

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17

Department:

Chemistry

Subject:

Organic chemistry

Semester:

6th

Submitted to:

Sir tadig shah

← long

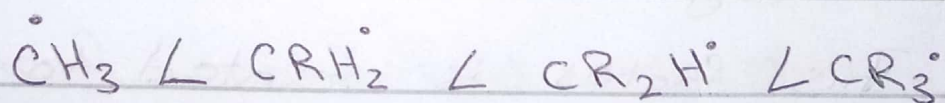
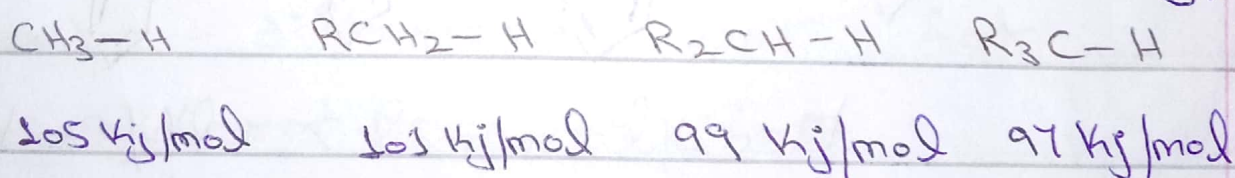
QV: Describe ~~two~~ <sup>five</sup> factors that affect the stability of free radicals?

Ans following factors affect the stability of free-radical:

1: Rate of substitution:

Radical stability is similar to the carbocation stability. Higher substituted radicals are more stable than less substituted radicals.

This can be proved on the basis of bond dissociation energy.



2: Resonance stability:

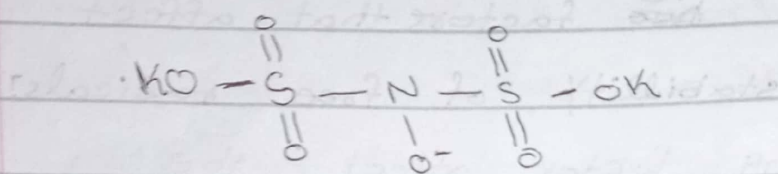
Higher the resonance stabilization. In other words, greater the delocalization of  $\pi$  electron in a free radical species, results in



greater stability e.g.  $\rightarrow$

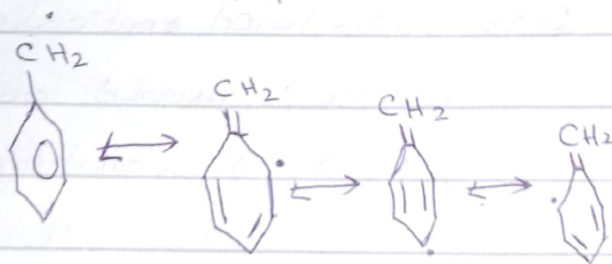
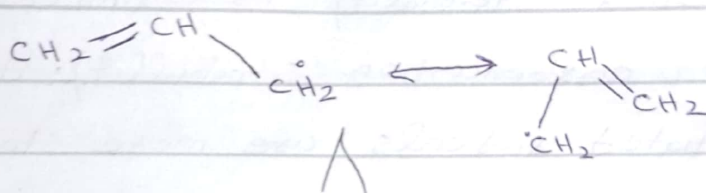
### 3: Steric hindrance:

Sometimes steric crowding around the center makes physically difficult for the radical to react with other molecules. e.g.



Potassium nitroso disulfonate

e.g.



Benzyl radical > alkyl radical

### 4: Hyperconjugation:

This factor is related to the overlap of adjacent  $\sigma$ -bonds with the P-orbital containing the unpaired electron in the free radical. When sigma bonds overlap with the P-orbital, it can stabilize the free

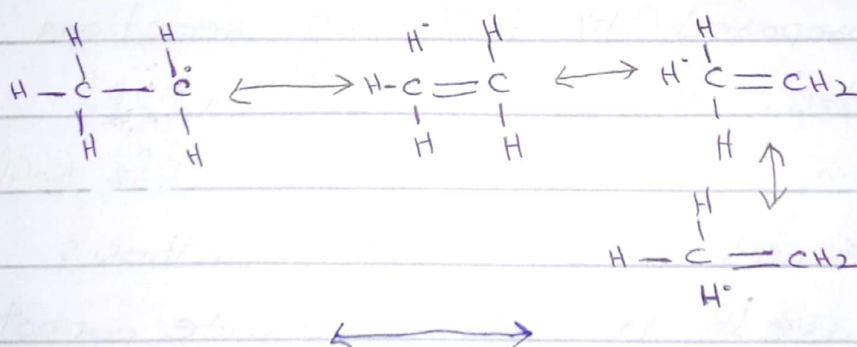
radical, making it less reactive and more stable. e.g.  $\rightarrow$

v: Electronic effects:

Electron donating groups can stabilize the free radical by donating electrons, making it more stable.

Electron withdrawing groups can destabilize the free radical by withdrawing electrons, making it more reactive.

e.g.  $\rightarrow$



v: Describe difference ways for the generation of free radical?

Ans Free radical produced by the cleavage of covalent bond. This cleavage requires sufficient amount of energy which is known as bond dissociation energy (BDE). Free radical are catalytic.



zed as follows:

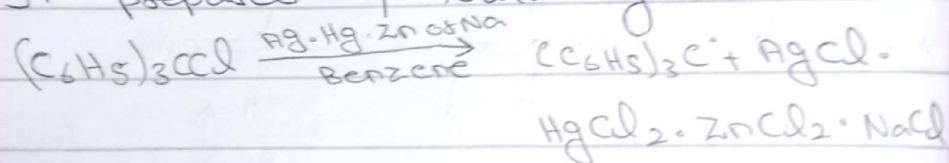
### 1: Stable/Long-lived free radicals:

Stable carbon radicals always contain extensive unsaturated conjugated  $\pi$  system, so that unpaired electron is free to spread throughout a large volume.

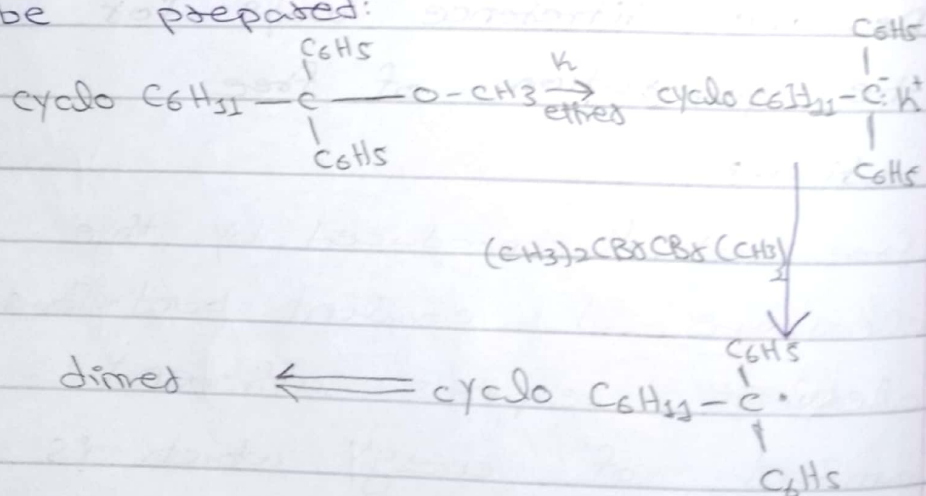
Following are different methods for the generation of free radicals.

(i) Tri-phenyl methyl was first stable free radical prepared by Gomberg.

It prepared by following reactions:

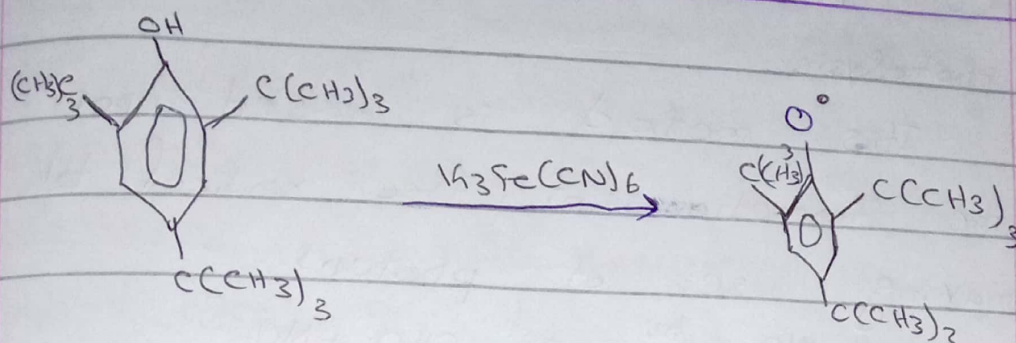


(ii) A second method of synthesis is used if related halide cannot be prepared:

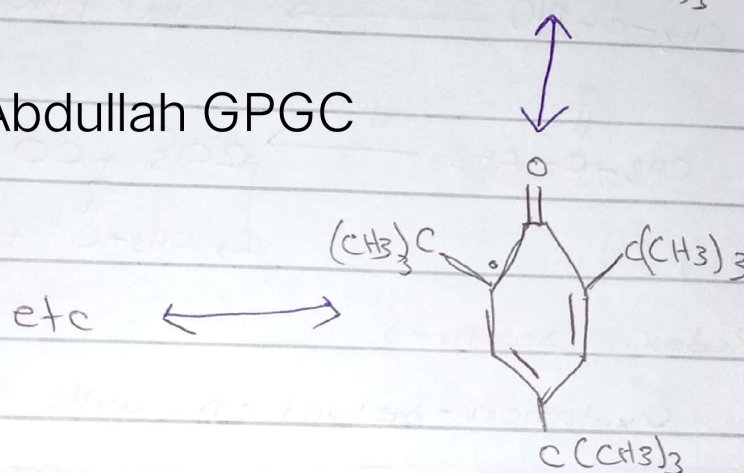


(iii) Highly substituted phenols, aryl amines and hydrazine after oxidation

produces free radicals.



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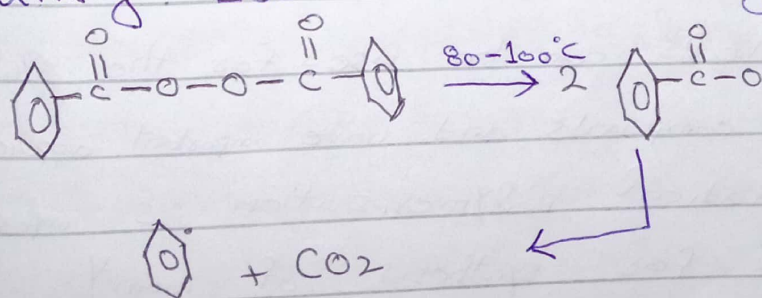


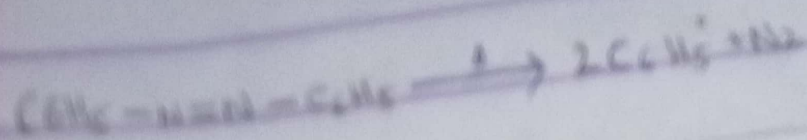
## 2: Short lived free radicals:

are highly reactive and difficult to handle. Three methods are used for the generation of short lived free radicals.

### (i) Thermolysis:

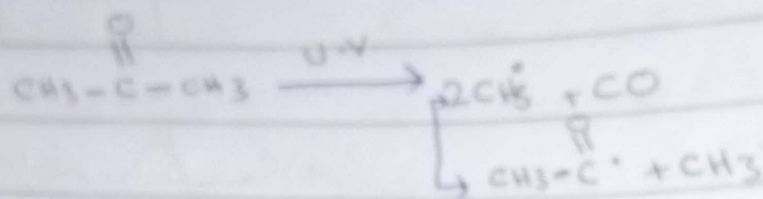
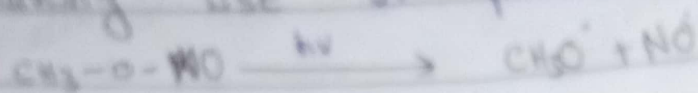
This method is based upon the formation of free radical by making use of heat energy.





(iii) Photolysis

This method is based upon the formation of free radical by making use of photons.



(iv) Redox reaction:

Oxidation-reduction with inorganic ions that can change their valence state by gain and loss of a single electron can be used for generation of free radicals.



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vi. write down the applications of free-radical.

Ans Free radicals can be used for the synthesis of many compounds and have important application.

(i) Free radical polymerization is important reaction for synthesis of many useful polymers like polystyrene,



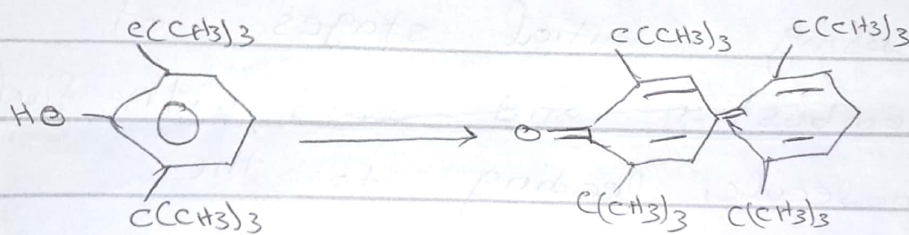
polyacetonitrile etc.

c) Many organic compounds can be synthesized by free radical mechanism. For example:

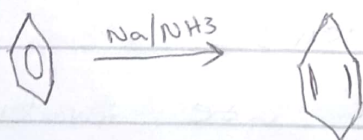
1) Synthesis of halo compounds

2) Kolbe's electrolysis for the synthesis of carboxylic acids

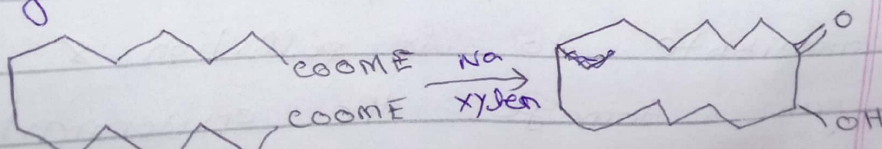
3) Oxidative coupling of phenols are used for synthesis of some dyes and drugs



4) Radical reduction reaction generally involves the addition of hydrogen to  $\pi$ -bond



5) The acyloin condensation is one of the best methods to use when synthesizing medium to large size ring.





(vi) Free radicals are produced during radiation treatment to damage cancer cells

Some drugs work by generating free radicals in specific tissues to target disease.

(vii) Free radical play a vital role in the combustion process. They are generated during initial stages of combustion and react with fuel molecule, leading to the release of energy in the form of heat and light

(viii) Free radicals can cause food spoilage by oxidizing fats & proteins.

(ix) Free radicals are involved in oxidative process of hair dyeing.

(x) Free radicals in atmosphere such as hydroxyl radicals contribute to air pollution & can damage the environment

(1)

Ans Radical scavengers are molecules or compounds that possess the ability to effectively quench and neutralize highly reactive free radicals. These scavengers are pivotal in protecting biological systems and materials from oxidative damage. Prominent radical scavengers include antioxidants such as vitamin C and vitamin E, which act by donating electrons or hydrogen atoms to free radicals, thereby preventing the chain reactions that lead to oxidative stress and damage.

Persistence refers to the ability of a free radical to maintain its existence over a period without reacting or undergoing significant changes. Some free radicals are more persistent, meaning they have a longer lifetime and are less likely to react immediately with other molecules. This persistence can depend on



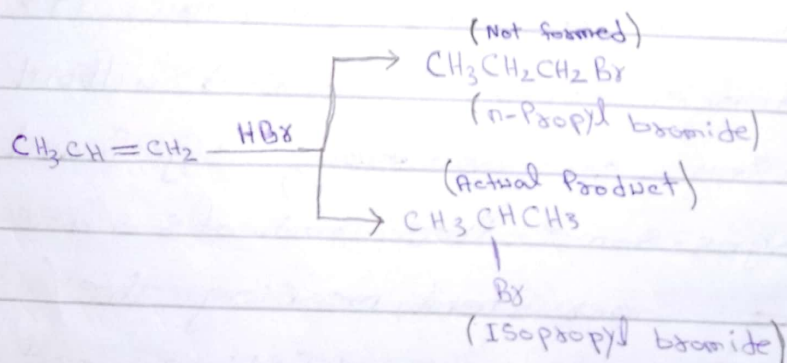
factors like the stability of the unpaired electron and environmental conditions e.g. hydroxyl radical are less persistent, so it is less stable. The ability of persistence is directly proportional to the unpaired electrons.



Q What are regioselective reactions?

Ans Reactions that can potentially yield two or more constitutional isomers but actually produce only one (or predominantly one) are known as regioselective reactions.

The regioselectivity is determined by steric hindrance. e.g.



(2)

Ans The Perch method for the detection of free radicals involves using

a chemical probe to trap and stabilize free radicals, making them easier to detect. The method is following

1: Chemical probe:-

A suitable chemical probe, often compound containing a specific functional group is chosen. This probe will react with and trap free radicals by forming a stable product.

2: Reaction:- Umer Abdullah GPGC

The chemical probe is introduced into the sample suspected to contain free radicals. When free radicals are present they react with probe, forming a stable ~~product~~ adduct.

3: Detection:-

The stable adduct formed as a result of the reaction is easier to detect using various analytical techniques such as spectroscopy or chromatography.

\* Example:

One common example of the



perch method is the use of 2,2,6,6-tetramethylpiperidine (TEMP) as chemical probe to detect superoxide radicals. In this case, TEMP reacts with superoxide radicals to form a stable product called TEMPO, which can be easily detected by electron paramagnetic resonance (EPR) spectroscopy.

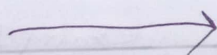


(3)

Ans Free radicals play a significant role in the aging process through a mechanism known as oxidative stress. Here's how it works:

#### 1: formation of free radicals:

Free radicals are highly reactive molecules with unpaired electrons. They are generated as by-products of normal metabolic processes in the body, such as energy production and immune responses.



2: Damage to cells and DNA:-

When free radicals are produced in excess or the body's antioxidants defenses are overwhelmed, they can damage various cellular components, including DNA, proteins, and lipids.

3: Accumulated Damage:-

Over time, the accumulated damage to cells and DNA can lead to mutations and dysfunctional cellular processes. This can result in the aging of tissues and organs, contributing to age-related diseases and declining.

4: Inflammation:-

Free radicals can also trigger inflammation, which is associated with the aging process. Chronic inflammation can cause further damage to cells and tissues.

5: Role in Age-related diseases:

Free radical-induced oxidative stress is linked to various age-related diseases, including



cardiovascular disease, neurodegenerative disorders like Alzheimer's and Parkinson's and certain types of cancer.

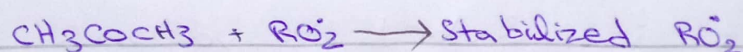
To mitigate the effects of free radicals and oxidative stress, it's essential to maintain a balanced diet rich in antioxidants such as Vitamin C and E.



Q What is captodative effect?

Ans Captodative effect is a phenomenon where the presence of a carbonyl group in a molecule contributes to the stabilization of adjacent free radicals. Specifically, the captodative effect involves the donation of electron density from the oxygen atom of the carbonyl group to the neighboring free radicals, which helps reduce their reactivity and enhances their stability.

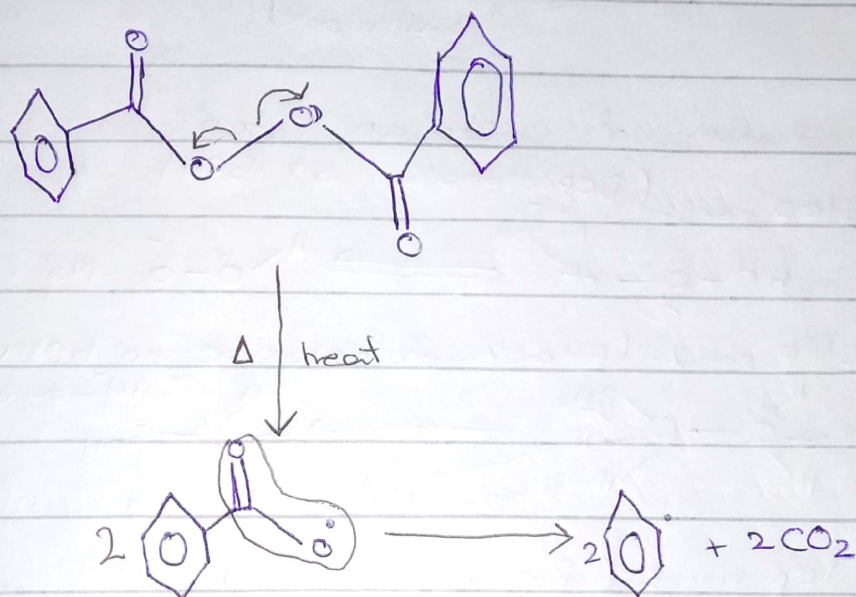
Example:-



This representation highlights the

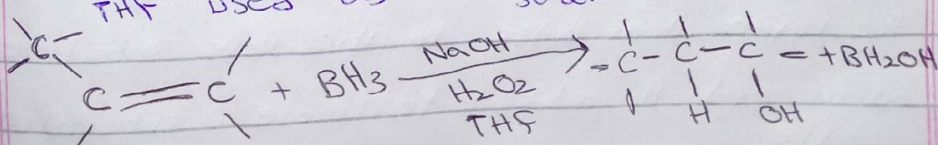
role of acetone in stabilizing the peroxy radical without specifying any further chemical transformation.

Q What product do you expect from thermal decomposition of benzoyl peroxide?



Q Describe briefly hydroboration with mechanism and regioselectivity?

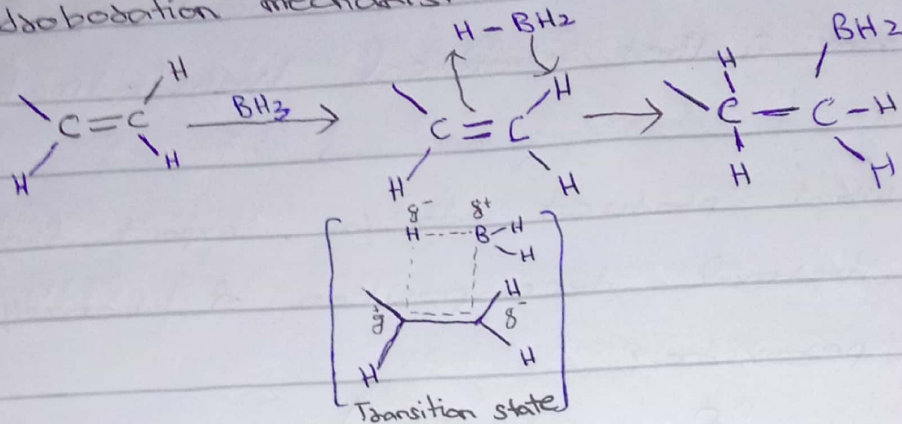
Ans Hydroboration involves the addition of  $\text{BH}_3$  to the double bond, with hydrogen become attached to one double carbon and boron to other. The alkyl borane can undergo oxidation, in which the boron is replaced by  $-\text{OH}$  group. THF used as a solvent.





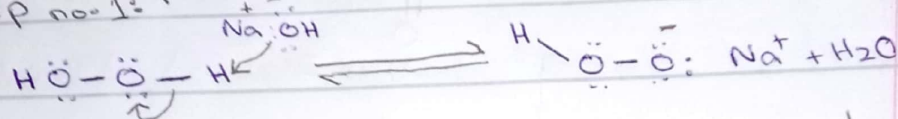
\* Mechanism:-

\* Hydroboration mechanism:-

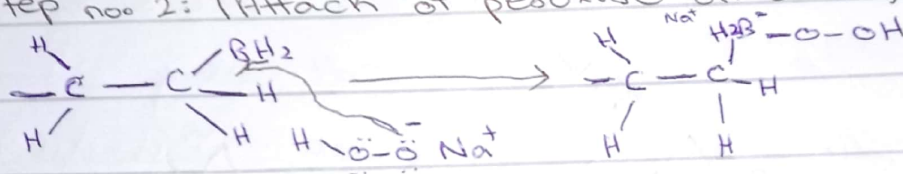


\* Mechanism of Oxidation step:-

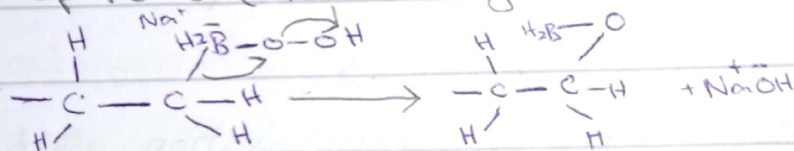
\* Step no. 1: (Deprotonation of  $H_2O_2$ )



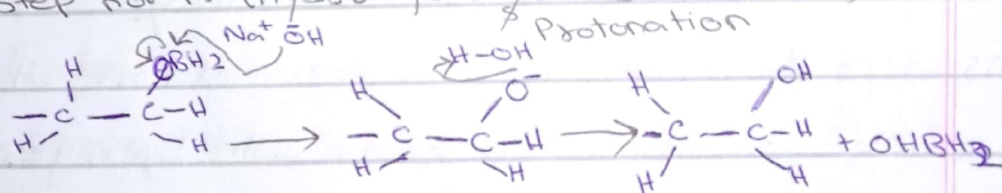
\* Step no. 2: (Attack of peroxide on boron)



\* Step no. 3: (Rearrangement)



\* Step no. 4: (Hydrolysis of B-O bond)



\* Regioselectivity:-

The formation of  $CH_3CH_2OH$  is result of anti-Markovnikov regioselectivity, where the boron atom adds to the less substituted atom and H adds to more substituted carbon.